Heliodoro Tejeda Lemus

Preferred Name: **Helio Tejeda**

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Heliodoro Tejeda Lemus Heliodoro Tejeda Lemus MADLAB at UCI

Education

2026 **Ph.D. Cognitive Sciences** **University of California, Irvine**
Dissertation title: *TBD*

2025 **M.Sc. Cognitive Neuroscience** **University of California, Irvine**
Thesis title: *From Collaborating with Machines to Integrating AI-Advice into Human Decision-Making: A Brief Review.*

2017 **B.S. Computer Science** **University of California, Irvine**
Specialization: *Intelligent Systems*
Minor: *Mathematics*

Employment History

2020 – Present **Graduate Student Researcher** **University of California, Irvine**

- Designed and implemented online behavioral experiments to investigate human–AI collaboration and decision-making processes.
- Examined how individuals integrate AI-advice into their judgments and choices using computational and statistical analysis methods.
- Contributed to the development of experimental paradigms, data collection pipelines, and analysis frameworks for large-scale online studies.
- Co-authored research manuscripts and presented findings at academic conferences and seminars.

2026 – 2026 **Instructor of Record** **University of California, Irvine**

- Designed and delivered a full undergraduate statistics course for 170 students, generating all course materials including lectures, homework assignments, quizzes, exams, and projects.
- Implemented interactive lessons that encouraged active student participation, including in-class problem solving and student-led demonstrations at the front of the classroom.
- Combined traditional lecture slides with live handwritten examples to illustrate step-by-step statistical reasoning and problem solving.
- Maintained comprehensive course resources by posting lecture slides, written notes, and recorded lectures to support student review and accessibility.
- Curated additional learning resources, including external videos and online materials, to provide students with multiple pathways for engaging with course concepts.

2022 – 2025 **Graduate Student Teaching Assistant** **University of California, Irvine**

- Assisted in the instruction of undergraduate courses, including preparing and delivering lectures, leading lab sessions, and facilitating class discussions.
- Provided guest lectures on specialized topics including Bias in AI, Programming Environment Setup, Monte Carlo Simulations, and Data Analysis & Visualization.
- Developed and graded assignments, quizzes, and examinations to assess student learning and progress.
- Held weekly office hours to provide academic support and mentorship to students.

Employment History (continued)

2017 – 2020 **Research Data Analyst** **Stanford University**

- Built an automated pipeline for the classification of cardiac malformations using CNN and RNN architectures.
- Created an internal webpage for the storage, viewing, and analysis of patient MRI data.

2016 – 2017 **Data Analyst Intern** **Restoration Media**

- Created python modules, connecting to third-party services and MySQL, to automate data processes.
- Designed and developed dashboard tools to monitor company wide data processes.

2015 – 2015 **Software Engineering Intern** **Matterport**

- Implemented scripts to collect machine data such as: CPU usage, Packages, Processes, Network Interfaces.
- Developed an internal dashboard from the ground up utilizing data visualization tools including: Highcharts, D3.js, Mixpanel.

2013 – 2015 **Trainer, Operations, & Driver** **Anteater Express**

- Responsible for ensuring that all drivers are proficient at performing their required duties.
- Served as a positive role model to all drivers and reported directly to Anteater Express Supervisors.
- Duties included: driving shuttles, performing bus maintenance, and managing small currency transactions.

2009 – 2012 **Life Guard & Swim Instructor** **City of Patterson**

- Instructed all age groups in swimming techniques providing a safe and comfortable swimming environment.

Professional Service

2025 – Present **National Liaison** **University of California, Irvine - Society for Advancing Chicanos/Hispanics & Native Americans in Science (SACNAS)**

- Served as the primary connection between the university SACNAS chapter and the national SACNAS organization.
- Promoted cross-campus collaboration by coordinating joint events and initiatives among regional SACNAS chapters.
- Expanded the local SACNAS community by engaging new members and fostering partnerships with student organizations and faculty allies.
- Communicated national SACNAS opportunities, including conferences, scholarships, and leadership programs, to chapter members.

2025, 2026 **Graduate Student Panelist** **University of California, Irvine - Cognitive Science Association (CSA)**

- Participated in annual graduate student panels advising undergraduate students about pursuing graduate school and research careers.
- Discussed topics including selecting graduate programs, deciding to pursue a PhD, developing effective study habits, and navigating daily life as a graduate student.
- Answered student questions about research training, academic career paths, and preparation for graduate-level study.

Professional Service (continued)

2025 **Panelist** **University of California, Irvine - NASPA Undergraduate Fellows Program (NUFP)**

- Participated in a panel discussion with NUFP undergraduate fellows about post-graduate pathways and research careers.
- Shared experiences about my path to graduate school and the process of identifying research interests in cognitive science and artificial intelligence.
- Provided guidance on navigating non-linear academic trajectories and preparing for graduate study.

2024 – 2025 **President** **University of California, Irvine - Society for Advancing Chicanos/Hispanics & Native Americans in Science (SACNAS)**

- Provided strategic leadership and oversight for both undergraduate and graduate SACNAS board members, coordinating joint initiatives and leadership development.
- Organized and facilitated meetings, workshops, and events supporting academic success, professional development, and community building among members.
- Strengthened partnerships with other student organizations and university offices to expand outreach and visibility of SACNAS initiatives.
- Fostered cross-campus collaboration by co-developing and hosting joint events that connected students, faculty, and researchers across departments.

2023 – 2024 **Social Events Chair** **University of California, Irvine - Society for Advancing Chicanos/Hispanics & Native Americans in Science (SACNAS)**

- Organized and hosted social and community-building events to strengthen connections among SACNAS members and the UCI community.
- Coordinated inclusive gatherings that fostered belonging, mentorship, and peer support among students from underrepresented backgrounds.
- Contributed to cultivating an inclusive and supportive community aligned with SACNAS's mission to advance diversity in science.

2022 – 2023 **Member** **University of California, Irvine - Society for Advancing Chicanos/Hispanics & Native Americans in Science (SACNAS)**

- Actively participated in SACNAS meetings, workshops, and community-building events supporting diversity and inclusion in STEM.
- Contributed to fostering an inclusive and supportive community for underrepresented students in science.

Mentoring Experience

2026 **Workshop Instructor IrvineHacks 2026** **University of California, Irvine**

- Delivered a workshop titled "Introduction to Cognitive Modeling" for hackathon participants interested in computational approaches to human behavior.
- Introduced core concepts of cognitive modeling and discussed connections to machine learning and artificial intelligence methods.
- Guided participants through a hands-on tutorial using Python in Google Colab from load behavioral data and implementing the AFD model.
- Adapted instructional materials from a computational modeling tutorial developed during my teaching assistantship in Computational Modeling and Behavior.

Mentoring Experience (continued)

2023 – Present **SACNAS Mentoria Mentor** **University of California, Irvine**

- Mentored undergraduate students from underrepresented backgrounds pursuing degrees in STEM fields.
- Provided guidance on academic development, research opportunities, and graduate school preparation.
- Supported mentees in developing scientific communication skills through feedback on presentations and writing.
- Shared insights on navigating academia, applying for research positions, and balancing personal and academic responsibilities.

2019 – 2020 **Lead Instructor** **Academe Vitae**

- Mentor elementary and middle school students about STEM.
- Teach JavaScript programming through playful learning and computer games.

Summer 2019 **Undergraduate Mentor** **Stanford University - Priest Lab**

- Mentored an undergraduate student in computer programming and machine learning.
- Guided them through converting cardiac MRIs into echocardiograms and vice versa.

Research Publications

Journal Articles

1. M. Steyvers et al., "What large language models know and what people think they know," *Nature Machine Intelligence*, vol. 7, no. 2, pp. 221–231, 2025.
2. M. Steyvers, **H. Tejeda**, G. Kerrigan, and P. Smyth, "Bayesian modeling of human–ai complementarity," *Proceedings of the National Academy of Sciences*, vol. 119, no. 11, e2111547119, 2022.
3. C. Tcheandjieu et al., "High heritability of ascending aortic diameter and trans-ancestry prediction of thoracic aortic disease," *Nature Genetics*, vol. 54, no. 6, pp. 772–782, 2022.
4. **H. Tejeda**, A. Kumar, P. Smyth, and M. Steyvers, "Ai-assisted decision-making: A cognitive modeling approach to infer latent reliance strategies," *Computational Brain & Behavior*, vol. 5, no. 4, pp. 491–508, 2022.
5. M. Yu et al., "Computational estimates of annular diameter reveal genetic determinants of mitral valve function and disease," *JCI insight*, vol. 7, no. 3, e146580, 2022.
6. A. Córdova-Palomera et al., "Cardiac imaging of aortic valve area from 34 287 uk biobank participants reveals novel genetic associations and shared genetic comorbidity with multiple disease phenotypes," *Circulation: Genomic and Precision Medicine*, vol. 13, no. 6, e003014, 2020.
7. J. A. Fries et al., "Weakly supervised classification of aortic valve malformations using unlabeled cardiac mri sequences," *Nature communications*, vol. 10, no. 1, p. 3111, 2019.

Conference Proceedings

1. **H. T. Lemus**, A. Kumar, and M. Steyvers, "An empirical investigation of reliance on ai-assistance in a noisy-image classification task.," in *HHAI*, 2022, pp. 225–237.

Books and Chapters

1. **H. Tejeda Lemus**, A. Kumar, and M. Steyvers, "How displaying ai confidence affects reliance and hybrid human-ai performance," in *HHAI 2023: Augmenting Human Intellect*, IOS Press, 2023, pp. 234–242.

Preprints

1. M. Steyvers et al. "The calibration gap between model and human confidence in large language models."
2. A. Córdova-Palomera et al. "Cardiac imaging of aortic valve area from 26,142 uk biobank participants reveal novel genetic associations and shared genetic comorbidity with multiple disease phenotypes."
3. C. Tcheandjieu et al. "High heritability of ascending aortic diameter and multi-ethnic prediction of thoracic aortic disease."
4. M. Yu et al. "Computational estimates of mitral annular diameter in systole and diastole cardiac cycle reveal novel genetic determinants of valve function and disease."
5. J. Shepard, A. Monji, and **H. Tejeda**. "Black box the ultimate game of hide and seek."

Skills

Languages	Strong reading, writing and speaking competencies for English and Spanish.
Coding	Python, MATLAB, Java, C/C++
Web Development	HTML/CSS, JavaScript, PHP, Django
Visualization Tools	Matplotlib, Seaborn, Highcharts, D3.js, Mixpanel
Databases	SQL, MySQL, PostgreSQL, SQLite, BigQuery
Misc.	Academic research, teaching, LaTeX typesetting and publishing.

Miscellaneous Experience

Certification

2024 **Center for the Integration of Research, Teaching, and Learning Associate Level**
UCI Division of Teaching Excellence and Innovation - Certificate of Completion

Coaching Experience

2019 – 2020 **Head Coach - Freshmen Soccer** **San Mateo High School**
- Served as the head coach for the freshmen soccer team at San Mateo High School.
- Continued to assist and work as the conditioning coach for the JV soccer team.

2018 – 2019 **Assistant Coach - Junior Varsity Soccer** **San Mateo High School**
- Served as the assistant coach for the JV soccer team at San Mateo High School.
- Won the division championship with an undefeated record of 14-0-1.

References

Available on Request